

Mohammad Ibrahim Memon

Robotics Engineer | Manipulation | Physical AI | ROS 2 | NVIDIA Isaac Sim

immemon1811@gmail.com | +49 15774001726 | Bonn, Germany

Portfolio | LinkedIn: ibrahimmemon18 | GitHub: ibrahim-1811



PROFESSIONAL SUMMARY

Robotics engineer and M.Sc. Autonomous Systems candidate focused on industrial robot manipulation, Physical AI, and digital twins. Experienced in developing ROS 2 and MoveIt 2 manipulation systems for real robots using NVIDIA Isaac Sim, RGB-D perception, motion planning, demonstration datasets, and imitation learning. Combines robotics software and robot-learning experience with hands-on embedded electronics, hardware integration, and system-level debugging.

EXPERIENCE

Robotics Intern | INVITE GmbH (Leverkusen, Germany)

05/2026 – Present

- Architected and integrated ROS 2, MoveIt 2, and NVIDIA Isaac Sim digital twins for UR10 and UR5e industrial workcells, including grippers, RGB-D cameras, TF, collision geometry, motion planning, and simulated sensor pipelines.
- Developed autonomous pharmaceutical manipulation workflows for bottle grasping, transport, liquid pouring, capping/decapping, scooping, and precision placement using perception-aware motion planning.
- Implemented a ROS 2–Isaac Sim control architecture integrating trajectory execution, joint-state feedback, simulation time, TF, RGB-D perception, and synchronization between MoveIt 2 and the simulated robot.
- Built an end-to-end demonstration acquisition pipeline for robot learning, recording synchronized robot states, actions, RGB, depth, object poses, task phases, and timestamps with automated episode quality validation and LeRobot dataset conversion.
- Developed and evaluated imitation-learning workflows using LeRobot and ACT, and implemented an Isaac Mimic pipeline for scalable manipulation demonstration generation.
- Conducted a feasibility study for pharmaceutical handling with the Unitree G1 humanoid, evaluating bimanual manipulation, reachability, balance, collision clearance, and flexible-bag handling using an NVIDIA Isaac Sim digital twin and physical mock-up.

Research Assistant | Garrulus (Bonn, Germany)

05/2025 – 12/2025

- Designed and validated PCB protection and power-distribution circuits for 3.3 V, 5 V, and 24 V rails, including solenoid drivers and eFuse-based protection.
- Developed power sequencing, inrush-current limiting, and reverse-polarity protection in collaboration with hardware and firmware teams for autonomous tree-seeding robots.

Research Assistant | Kelo Robotics (Stuttgart, Germany)

05/2024 – 08/2024

- Built and evaluated NVIDIA Isaac Sim and Gazebo simulation environments for robotic navigation and control experiments.
- Compared simulator capabilities for multi-agent robotics, including sensor integration, control interfaces, simulation fidelity, and ROS integration.

Robotics System Engineer | Unbox Robotics Pvt. Ltd (Pune, India)

06/2022 – 07/2023

- Developed automated validation and debugging frameworks for autonomous warehouse robots, supporting feature verification, regression testing, and deployment readiness.
- Performed root-cause analysis, DFMEA, control-board debugging, and oscilloscope-based electrical validation to diagnose system-level failures.
- Coordinated cross-functional integration, robot deployments, and customer demonstrations while serving as Scrum Master.

Embedded System Engineer | DashDot Robotics Pvt. Ltd (Ahmedabad, India)

03/2021 – 02/2022

- Designed and prototyped embedded hardware and sensor interfaces for robotic systems, from schematic-level development through hardware bring-up.
- Performed circuit debugging, root-cause analysis, sensor calibration, and firmware integration across I2C, SPI, and UART interfaces.

RESEARCH & PUBLICATIONS

M. I. Memon, U. Patil, I. Awaad, Y. M. Youssef, "Learning Interpretable Scheduling Policies for Online Multi-Robot Task Allocation via Inductive Logic Programming," **PlanRob, ICAPS 2026**, Dublin, Ireland, 2026.

Patent: "An Inventory Management, Tracking and Consumption Predicting System using Intelligent Container Box and Central Server," Indian Patent Publication No. IN 202021029069 A, 2020.

SKILLS

Programming : Python, C++, C, Bash, MATLAB

Robotics : ROS 2, MoveIt 2, tf2, RViz, URDF/Xacro, OMPL

Simulation & Digital Twins : NVIDIA Isaac Sim, Isaac Lab, Isaac Mimic, USD, PhysX, Gazebo

Robot Learning : PyTorch, LeRobot, ACT, Imitation Learning, Demonstration Collection, Dataset Validation, Policy Evaluation

Perception : RGB-D Cameras, OpenCV, YOLO

Embedded & Electronics : PCB Design, Power Electronics, I2C, SPI, UART, Hardware Bring-up, Oscilloscope Debugging, eFuse Protection

Engineering Tools : Linux, Git, CI/CD, Integration Testing, Root Cause Analysis, DFMEA

PROJECTS

Intrinsic AI for Industry Challenge | ROS 2, YOLO, PyTorch, LeRobot, ACT

- Ranked **Top 40 of 400 teams** in the Intrinsic AI for Industry Challenge for a contact-rich cable and connector manipulation task.
- Built a multi-camera perception and demonstration pipeline using **YOLO-based object detection**, robot state, and task-state information for manipulation and dataset validation.
- Collected and validated **157 successful demonstrations**, converted the data to LeRobot format, and trained an **ACT policy for 100,000 steps** for learned manipulation and insertion.

Robothon 2025 (Grand Challenge) | ROS, Python, C++, MoveIt, Kinova Gen3

- Placed **Top 5** in the international Robothon 2025 Grand Challenge by developing autonomous manipulation solutions under strict task and time constraints.
- Built a vision-guided manipulation pipeline using MoveIt and OpenCV for task-state detection, button pressing, stylus grasping, and trajectory drawing.
- Implemented a maze-solving routine using a magnetic end-effector with continuous visual feedback for path correction.

ILP-Based Fault Detection and Diagnosis for Mobile and Multi-Robot Systems | ROS, Python, Prolog, ILP

- Developed an interpretable fault detection and diagnosis framework for ROS-based mobile robots using Inductive Logic Programming.
- Modelled faults including wheel slip, sensor noise, delay, and communication inconsistencies using symbolic rules.
- Evaluated the approach in simulation and real robot experiments, and extended it to multi-robot scenarios to study fault propagation and coordination robustness.

Explainable Multi-Robot Task Allocation | ROS 2, Python, Prolog, ILP, A*

- Developed an online multi-robot task-allocation framework for warehouse environments combining grid-based A* path planning, scheduling, and symbolic robot/task state modelling.
- Benchmarked heuristic, auction-based, optimization, and learning-based allocation strategies using task completion, travel, energy, congestion, and scheduling objectives.
- Developed interpretable scheduling policies using Inductive Logic Programming, leading to a publication at PlanRob, ICAPS 2026.

EDUCATION

Hochschule Bonn-Rhein-Sieg
U V Patel College of Engineering

M.Sc. Autonomous Systems — Bonn, Germany
B.Tech. Mechatronics Engineering — India

Expected: 06/2027
2018 – 2022

LANGUAGES

English : C1

German : A1 (currently learning)